

Modern Application Development II Project Report

1. Student Details

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About Me: I am pursuing my last term at diploma level. I am interested in machine learning and solving real world problems by applying the knowledge I gain in studying and practicing in this degree.

2. Project Details

Project Title: Evertrek – A Trekking Management Application

Problem Statement:

Trekking organizations often manage treks, staff assignments, participant registrations, available slots, and booking records manually through spreadsheets, calls, or messages, which can lead to errors and poor coordination. Required to build a centralized web application where Admins, Trek Staff, and Trekkers can securely manage trekking operations, bookings, participant status, trek availability, reports, reminders, and trekking history.

Approach:

The app was built following a modular role-based approach by developing separate Flask API blueprints and Vue interfaces for Admin, Trek Staff, and Trekker users. JWT authentication and backend validation protect operations, while SQLite stores data and Redis/Celery handle caching, OTP emails, reminders, reports, and CSV exports. The app connects the frontend through Axios, the app has been tested for each role's workflow and it's been for reliable local execution and easy maintenance.

3. AI/LLM Declaration

I used **ChatGPT 5** to assist in finding bugs, UI and UX suggestions and improvements. Writing better documentation, comments and best practices.

The extent of AI/LLM usage is around 20%, intended to bug finding, code suggestions, documentation and formatting

4. Technologies and Frameworks Used

Technology / Library	Purpose
Python	Core backend programming language
Flask	Backend web framework for building REST APIs
SQLAlchemy	Object-relational mapper for SQLite database operations
SQLite	Lightweight database for storing users, treks, bookings, and staff data
Vue.js 3	Frontend framework for building the user interface
Vue Router	Client-side routing and role-based page navigation
Bootstrap 5	Responsive frontend styling and layout
Axios	Communication between the Vue frontend and Flask API
JWT	Authentication and role-based access control
Redis	OTP storage, caching, and Celery message broker
Celery	Background processing for emails, reminders, reports, and CSV exports
Celery Beat	Scheduling daily and monthly background jobs
Gmail SMTP	Sending OTP, reminder, report, and export emails
Vite	Frontend development server and build tool
Werkzeug	Password hashing and security utilities
Git and GitHub	Version control and project management

5. Database Schema / ER Diagram

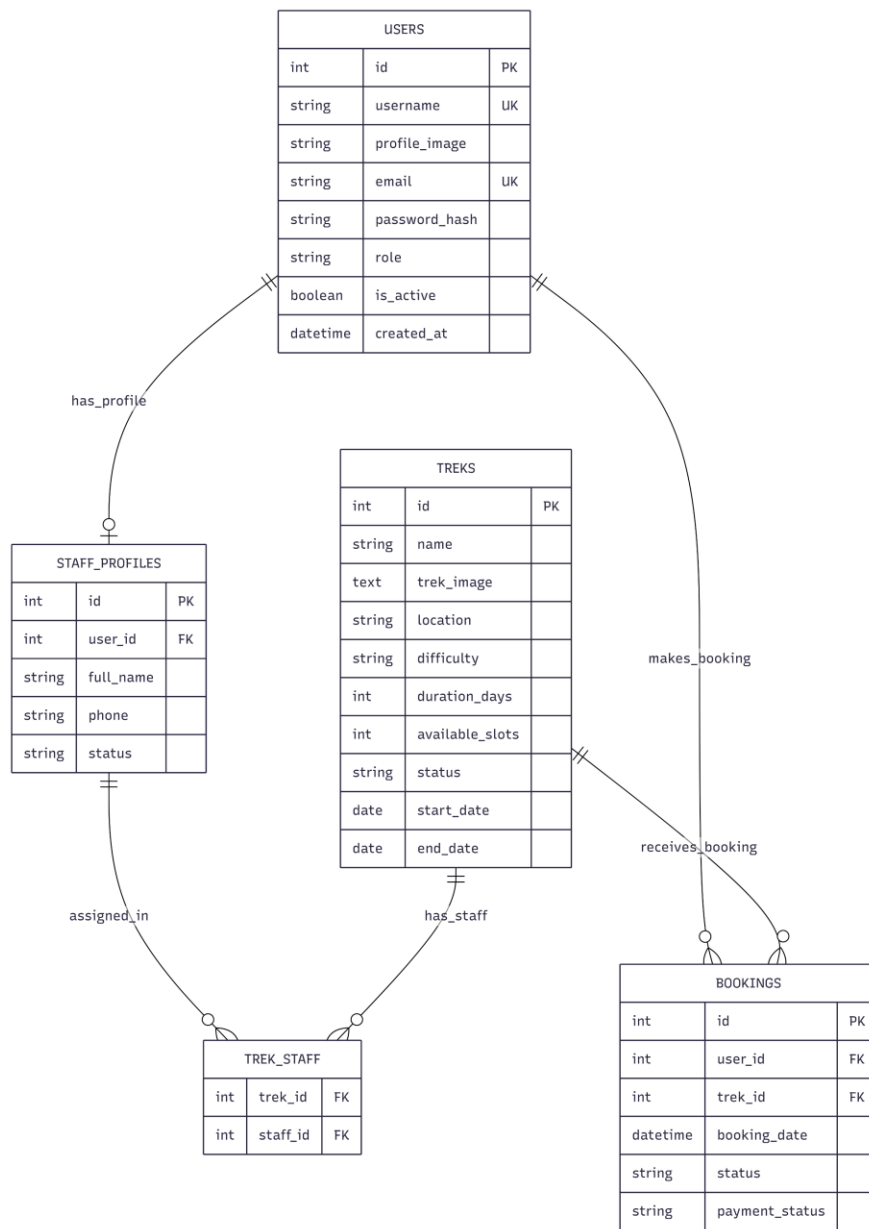
Tables:

1. User — stores Admin, Staff, and Trekker account details (id, username, email, password_hash, role, is_active)
2. StaffProfile — Stores additional staff details (id, user_id, full_name, phone, status)
3. Trek — stores trekking route details (id, name, location, difficulty, duration_days, available_slots, status, dates)
4. Booking — stores trek booking details (id, user_id, trek_id, booking_date, status, payment_status)
5. Trek Staff Association — connects staff members with assigned treks (trek_id, staff_id)

Relationships:

- One-to-One → User → StaffProfile
- One-to-Many → User → Booking
- One-to-Many → Trek → Booking
- Many-to-Many → StaffProfile ↔ Trek through Trek Staff Association

The role field in the User table distinguishes between Admin, Staff, and Trekker accounts.



6. API Resource Endpoints

There are total 36 API routes included separately in the submission ZIP as **api.yaml**

Endpoint	Method	Description
/api/auth/login	POST	Authenticate Admin, Staff, or Trekker and return JWT token
/api/auth/register	POST	Start Trekker registration and send OTP email
/api/auth/verify-otp	POST	Verify OTP and create Trekker account
/api/public/treks/	GET	Fetch treks for public landing page
/api/admin/dashboard/	GET	Fetch admin dashboard statistics
/api/admin/treks/	GET	Fetch all treks

7. Architecture and Features

Architecture Overview:

- **backend/app.py** – main Flask application entry point and blueprint registration
- **backend/models.py** – SQLAlchemy database models for users, staff, treks, bookings, and staff assignments
- **backend/routes/** – Flask API blueprints for auth, admin, staff, trekker, and public routes
- **backend/tasks/** – Celery background tasks for emails, reminders, reports, and CSV exports
- **backend/extensions.py** – centralized setup for database, JWT, Redis, and CORS
- **frontend/src/** – Vue frontend application with router, store, Axios API layer, views, and components
- **SQLite** – local database used for storing users, treks, staff profiles, and bookings
- **Redis + Celery** – used for caching, OTP storage, background jobs, and scheduled tasks

Implemented Features:

- User registration and login with OTP verification
- JWT-based authentication and role-based access control
- Admin dashboard with platform statistics
- Trek creation, update, deletion, and status management
- Staff account creation, activation, blacklisting, and trek assignment
- Trekker management with search, profile view, and blacklist/activate controls
- Staff dashboard for assigned treks, slot updates, trek status updates, and participant management
- Public trek discovery page for browsing available treks
- Trekker booking system with duplicate booking prevention and slot validation
- Booking history, cancellation, and CSV export request
- Redis caching for frequently used API data
- Celery tasks for OTP email, staff credential email, reminders, reports, and exports
- HTML email system using Gmail SMTP

Additional Features:

- Public landing page for trek discovery before login
- Role-specific layouts for Admin, Staff, and Trekker users
- Dark/light theme support in the frontend
- CSV booking history export sent through email
- Daily trek reminder emails using Celery Beat
- Monthly/admin report generation support
- Responsive Bootstrap 5 UI with Axios-based API communication

8. Video Presentation

Google Drive Link: <https://drive.google.com/file/d/1uzpVcEyPwWlurLzLLokW16ss7XziX1vo/view?usp=sharing>